Breast Cancer Image Classification: A Comparative Study of SVM, CNN, and Transformer Models on the CBIS-DDSM Dataset

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Abstract

This study aims to evaluate and compare three machine learning approaches for breast cancer classification using mammogram images from the CBIS-DDSM: Breast Cancer Image Dataset, the largest publicly available dataset for breast cancer diagnosis research. The dataset includes 2,620 high-quality labeled mammograms categorized as normal, benign, and malignant, with verified pathological annotations. This enables a robust assessment of the methods explored in three selected research papers. Feature preprocessing is a crucial step in machine learning workflows, as it ensures data consistency, enhances model performance, and enables effective learning. Among preprocessing techniques, feature normalization is particularly important, as it standardizes feature scales to prevent bias in classical algorithms like Support Vector Machines (SVM) and Random Forest, which are sensitive to feature magnitude. This study explores the impact of feature preprocessing and normalization on breast cancer image data by leveraging both classical machine learning models and advanced deep learning architectures. We conduct experiments using multiple views of breast cancer images to capture diverse patterns and evaluate the performance of these methods. For classical models, feature selection and normalization significantly improve predictive accuracy, while for deep learning, we employ a residual convolutional neural network (ResNet) architecture to achieve robust feature extraction and classification. Results highlight the complementary strengths of classical and deep learning approaches in medical image analysis, providing insights into their integration for more accurate and reliable breast cancer diagnosis.

1 Introduction

The proposed method employs the Gray-Level Co-Occurrence Matrix (GLCM) for texture-based feature extraction in conjunction with a Support Vector Machine (SVM) classifier, focusing on the impact of different GLCM angles for optimal feature representation. The second approach utilizes an EfficientNet-based convolutional neural network (CNN) combined with a multi-stage transfer learning strategy to develop a "two-view classifier," analyzing both craniocaudal and mediolateral oblique views of mammograms. The third method leverages residual neural network architecture, improving classification accuracy through enhanced information sharing across views.

By implementing these three approaches, this study provides a comparative analysis of traditional machine learning techniques, such as SVMs, and modern deep learning frameworks, including CNNs and residual neural networks, to evaluate their effectiveness in breast cancer image classification. The findings aim to contribute insights into the evolving landscape of computational tools for medical image analysis.

2 Dataset

The Curated Breast Imaging Subset of the Digital Database for Screening Mammography (CBIS-DDSM) is a standardized and enhanced version of the original DDSM, created to address challenges in developing and benchmarking decision support systems for breast cancer diagnosis in mammography. The CBIS-DDSM includes 2,620 mammography studies encompassing normal, benign, and malignant cases with verified pathology information. Curated by trained mammographers, the dataset provides updated Region of Interest (ROI) segmentation, bounding boxes, and pathological diagnoses, with images decompressed and converted to the modern DICOM format. Unlike the original DDSM, which was hindered by outdated compression formats and imprecise ROI annotations, CBIS-DDSM ensures consistency and replicability, enabling robust evaluations of CADx (diagnosis) and CADe (detection) algorithms. The collection addresses historical limitations in public datasets, such as size, accessibility, and annotation precision, offering a reliable benchmark for the mammography research community. Metadata structure in CBIS-DDSM accounts for multiple patient IDs per participant, creating a discrepancy in the perceived participant count, but accurately includes 1,566 participants. Further details on the dataset's usage are available in the accompanying manuscript (Nature Scientific Data). For inquiries, contact Dr. Daniel Rubin, Stanford University (dlrubin@stanford.edu).(1)

3 Methodology

3.1 Impletation 1

3.1.1 Data Pre-processing

In this study, the Calc-Training Full Mammogram Images (DICOM) Dataset was utilized, comprising 1,546 breast cancer ROI cases categorized into three classes: BENIGN (528 cases), MALIGNANT (544 cases), and BENIGN WITHOUT CALLBACK (474 cases). The BENIGN WITHOUT CALLBACK category refers to identified benign cases that required no additional films or biopsy for confirmation. Since these cases lack further diagnostic follow-up, they were excluded from the training and testing phases.

The dataset was split into 70% training data and 30% testing data. Specifically, the training set included 352 BENIGN cases and 363 MALIGNANT cases, while the test set comprised 176 BENIGN cases and 181 MALIGNANT cases. This division ensures a balanced and representative dataset for evaluating the model's performance in classifying breast cancer ROI.

Before preprocessing, the data is converted from DICOM format to PNG format. The preprocessing begins with a grayscale conversion, where the RGB values of the images are equalized. Next, contrast enhancement is performed to distinguish the background from the foreground. This contrast sharpening is achieved using histogram equalization.

The Gray-Level Co-Occurrence Matrix (GLCM) method is utilized during the feature extraction stage. GLCM is a technique that captures spatial relationships between pixels by constructing a co-occurrence matrix based on pixel interactions at four specific angles. Determining the optimal angle for GLCM is crucial for accurate feature extraction. Finally, a Support Vector Machine (SVM) is employed to classify mammogram regions of interest (ROI) as benign or malignant.

3.2 Feature Normalization

Feature normalization plays a pivotal role in ensuring that machine learning models perform optimally, especially when the features have varying ranges and distributions. In this section, we describe and analyze the impact of different normalization methods on Gray Level Co-occurrence Matrix (GLCM) features based on our experiments.

3.2.1 Baseline Test Without Normalizing GLCM Features

To establish a baseline, we evaluated the model's performance using raw GLCM features without applying any normalization. The results revealed significant inconsistencies, with the model showing sensitivity to features with larger magnitudes. This was particularly evident in classical algorithms like Support Vector Machines (SVM), which rely on the calculation of distance-based margins, and

Random Forest, where unnormalized features caused certain splits to dominate the decision-making process.

3.2.2 Standard Scaling

Standard scaling was applied to the GLCM features to transform them into a standardized format with a mean of zero and a standard deviation of one. This scaling ensures that all features contribute equally to the model. The formula used for standard scaling is:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where x is the original feature value, μ is the mean of the feature, and σ is the standard deviation. After applying standard scaling, the model showed a marked improvement in stability and accuracy. SVM, in particular, benefitted significantly as the scaled features helped it define hyperplanes more effectively. Random Forest also showed enhanced performance due to more balanced feature importance distributions.

3.2.3 Power Transformer

To address potential skewness in the GLCM features, we applied the Power Transformer, which maps the features to a more Gaussian-like distribution. This transformation is especially useful for features with highly skewed distributions. The Power Transformer was implemented with the following transformation:

$$x_{\text{transformed}} = \begin{cases} \frac{(x+1)^{\lambda}-1}{\lambda}, & \text{if } \lambda \neq 0 \\ \log(x+1), & \text{if } \lambda = 0 \end{cases} \tag{2}$$

The transformed features were less skewed, improving the robustness of both SVM and Random Forest models. The Power Transformer also contributed to better convergence in models requiring gradient-based optimization, like logistic regression.

3.2.4 Yeo-Johnson Method for Normalization

Based on the outcomes of the previous experiments, we decided to produce the final normalized features using the Yeo-Johnson transformation, a variation of the Box-Cox transformation that can handle both positive and negative feature values. The Yeo-Johnson transformation is defined as:

$$x_{\text{transformed}} = \begin{cases} \frac{[(x+1)^{\lambda}-1]}{\lambda}, & \text{if } \lambda \neq 0, x \geq 0 \\ \log(x+1), & \text{if } \lambda = 0, x \geq 0 \\ \frac{-[(|x|+1)^{2-\lambda}-1]}{\lambda}, & \text{if } \lambda \neq 2, x < 0 \\ -\log(|x|+1), & \text{if } \lambda = 2, x < 0 \end{cases} \tag{3}$$

This method ensured that all features, regardless of their original distribution or sign, were normalized effectively. Applying Yeo-Johnson normalization resulted in the most consistent and improved performance across all models. SVM achieved optimal hyperplane separation, and Random Forest exhibited balanced tree structures with evenly distributed splits.

3.2.5 Model 1: Support Vector Machines

Support Vector Machine (SVM) is employed to classify mammogram images into benign and malignant categories. Four feature sets, as calculated in GLCM, serve as the input vector for the SVM. The penalty parameter of the SVM is set to 1, and the radial basis function (RBF) kernel is used. The mathematical equation for the RBF kernel is provided in Equation (1), where γ (Gamma) is calculated as shown in Equation (2). The value of γ in the RBF kernel must be greater than 0. The SVM training process terminates when the error reduction is less than 0.001.

$$rbf = \exp\left(-\gamma \|x - x'\|^2 \gamma\right) \tag{4}$$

$$\gamma = \frac{1}{\text{number of features}} \tag{5}$$

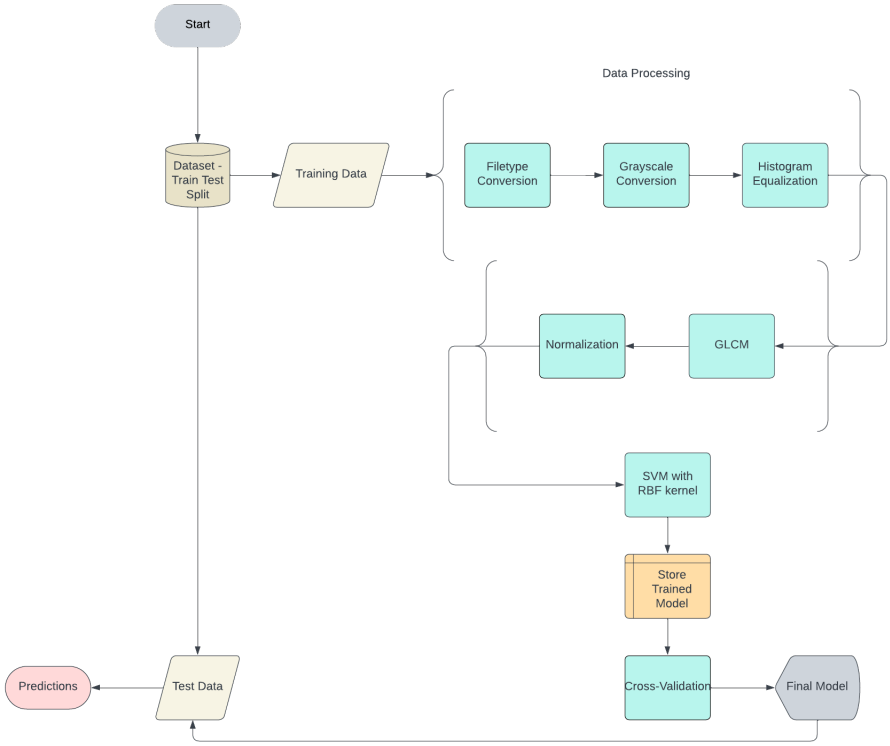


Figure 1: Process workflow for the implementation 1

3.3 Implementation 2

3.3.1 Data Augmentation

Data augmentation is a technique used to artificially increase the size and diversity of a dataset by applying various transformations to the existing data. It is particularly valuable in scenarios with limited labeled data, such as medical imaging, where acquiring new samples can be expensive or impractical. By introducing variability in the data, augmentation helps models generalize better and reduces the risk of overfitting. As part of the preprocessing, all mammograms were resized to 1152×896 pixels to address GPU memory limitations. The mean pixel value of all training images was subtracted from the training, validation, and test sets. Data augmentation was applied during all training processes in this study, with the following parameters: rotation of ±25°, zoom of ±20%, shear of ±12%, intensity shift of ±20%, and horizontal/vertical flips. Border reflection was used to fill areas outside the image domain.

3.3.2 Model 2: EfficientNet-Based Convolutional Network

Convolutional Neural Networks (CNNs) are a type of deep learning model designed to process structured grid data, such as images, by leveraging their spatial structure. They consist of convolutional layers that apply filters (kernels) to extract features like edges, textures, and patterns, pooling layers that reduce the spatial dimensions of feature maps while retaining important information, and activation functions like ReLU to introduce non-linearity for learning complex patterns. Fully connected layers are used at the end of the network for classification tasks, and techniques like dropout are applied to prevent overfitting. Advanced architectures, such as ResNet, incorporate residual connections to enable deeper networks and address the vanishing gradient problem. CNNs

process images hierarchically, with early layers learning basic features (e.g., edges) and deeper layers capturing abstract patterns (e.g., shapes). These networks are widely used in applications such as medical imaging, object recognition, facial recognition, and autonomous driving, due to their ability to effectively identify and process spatial patterns in high-dimensional data. EfficientNet is a family of Convolutional Neural Network (CNN) architectures designed to achieve high accuracy while being computationally efficient, introduced by Google in 2019. Its core innovation lies in a novel compound scaling method that systematically scales the network's depth, width, and resolution in a balanced manner to optimize performance without unnecessary computational costs. The base model, EfficientNet-B0, was created using Neural Architecture Search (NAS) to balance accuracy and efficiency, and larger variants (EfficientNet-B1 to B7) are scaled versions of this baseline. The architecture incorporates Mobile Inverted Bottleneck Convolution (MBConv) blocks and depthwise separable convolutions to reduce computation while maintaining expressive power. Squeeze-and-Excitation (SE) blocks are also used to recalibrate channel-wise features, enhancing the network's representational capacity. EfficientNet models outperform traditional architectures like ResNet and Inception in accuracy with fewer parameters and lower computational demands, achieving state-of-the-art results on benchmarks like ImageNet. These models are widely applied in tasks such as image classification, object detection, and medical imaging, where efficiency and performance are paramount.

In standard mammography, each breast is imaged in two views: craniocaudal (CC) and mediolateral oblique (MLO), resulting in abnormalities appearing in both views. I implement a convolutional neural network that simultaneously processes both views of the same breast, leveraging transfer learning for enhanced performance. The two-view classifier is built upon the weights of a single-view classifier, and the entire system is trained end-to-end. Additionally, we evaluated various combinations of the number of blocks and strides to determine the optimal network architecture.

3.4 Model 3: Residual Neural Network

A Residual Neural Network (ResNet) is a deep learning architecture as shown in Figure 2 designed to address the vanishing gradient problem that occurs in very deep networks, enabling the training of significantly deeper models. Introduced by Microsoft in 2015, ResNet uses a unique structure called residual blocks, where the input to a block is added directly to its output, creating a "shortcut" connection. This shortcut allows the network to learn residual mappings rather than the complete transformation, making optimization easier and preventing degradation of performance in deeper networks. The mathematical formulation involves learning the residual function $F(x) = H(x) - x$, where $H(x)$ is the desired mapping and x is the input. If $F(x)$ becomes zero, the block effectively behaves as an identity mapping, ensuring that deeper layers do not negatively impact the network's performance. ResNet models, such as ResNet-50 and ResNet-101, have demonstrated groundbreaking results in image recognition tasks and serve as the foundation for many modern architectures, offering superior accuracy and generalization across diverse applications.

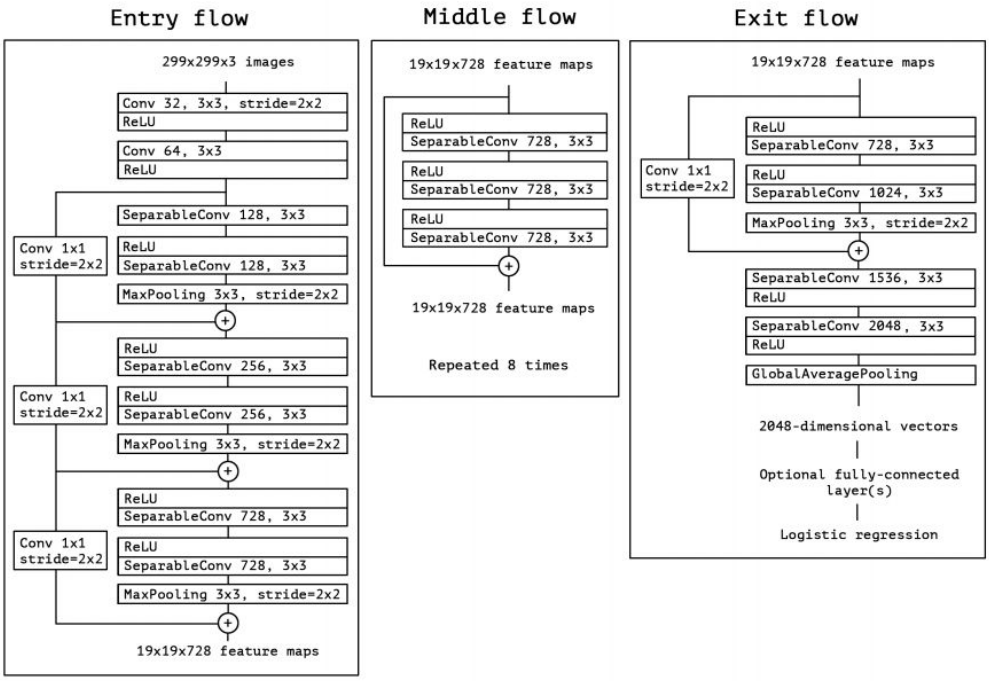


Figure 2: Architecture of Residual Neural Network - Xception

4 Evaluation

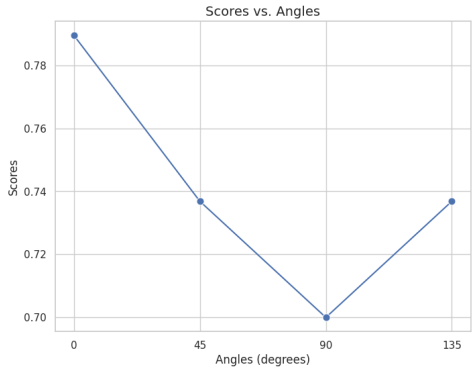
To evaluate the Support Vector Machine, we run the model on our test data and verify its performance based on the accuracy of classification. Initially, we tried to reproduce the Support Vector machine configuration provided by the authors of cite with $\theta = 45degrees$ for GLCM features. The results are shown in the following table 1.

Class	Precision	Recall	F1-Score	Support
BENIGN	0.30	0.05	0.09	140
MALIGNANT	0.51	0.90	0.65	157
Accuracy			0.50	297
Macro Avg	0.41	0.47	0.37	297
Weighted Avg	0.42	0.50	0.39	297

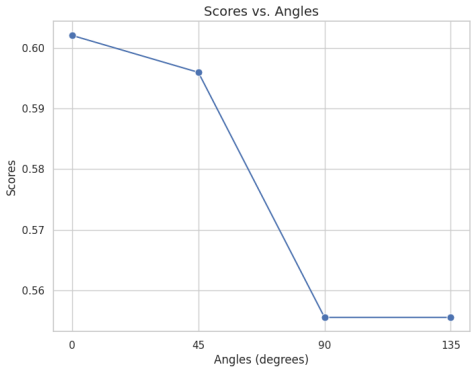
Table 1: Performance of SVM model with cross-validation ($angle = 45\,\mathrm{deg}$)

The reproduced result varies significantly from the scores mentioned in the paper (2). The possible explanation for this would be a mismatch of samples in the dataset. Although the model performed poorly in the experiment mentioned before, we will show that it can be improved significantly by label encoding, feature normalization, and cross-validation.

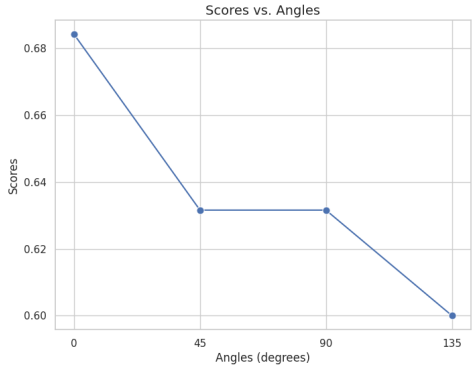
As shown in the figure 3 below, we ran the model on our test data and tested different feature normalization techniques. Using the Power Transformer (Yeo-Johnson method), the model achieved the highest accuracy (above 0.7) at $\theta = 45^\circ$ and $\theta = 135^\circ$, with a sharp drop at $\theta = 90^\circ$. The Standard Scaler showed a similar trend but slightly lower accuracy overall. Without normalization, accuracy remained below 0.68, demonstrating the importance of preprocessing. The 10-fold cross-validation without normalization performed worst, with accuracy dropping below 0.4 at higher angles. These results highlight that normalization techniques are critical for improving classification accuracy, especially at smaller angles.



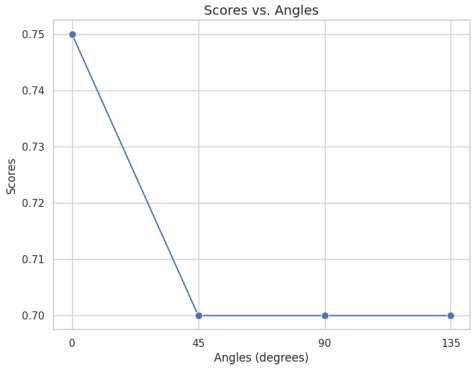
(a) Cross-validated accuracy of SVM using Power Transformer (Yeo-Johnson method) as feature normalization technique.



(b) 10-fold Cross-validated accuracy.



(c) Cross-validated accuracy of SVM without using any feature normalization technique.



(d) Cross-validated accuracy of SVM using standard scaler as feature normalization technique.

Figure 3: (a), (b), (c), and (d) show results of SVM using different techniques.

Similarly, reproduced the 2-views classifier and verified the performance with respect to test images from CBIS-DDSM dataset as depicted on Figure 4. We also incorporate the two-views approach from (3) and tried to simulate the classification behavior. Although, it generates proper output and classifies the sample well, it proved to be a tedious task to accomplish with the limited computation resources I have at my service. So to prove the applicability of neural networks I use the residual neural network architecture mentioned in (4). In order to analyze the impact of hyperparameter tuning on model performance, I experimented with different configurations, which are detailed in the table 2. We use Optuna to identify the best parameters for our use case.

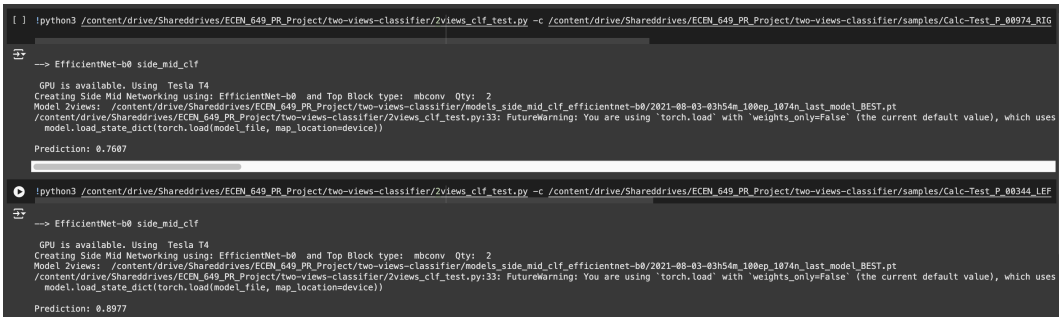


Figure 4: Classification results on test images by EfficientNet-Based Convolutional Network

Config	Changes	Accuracy		Val Accuracy	
		Left	Right	Left	Right
Config1	Learning rate: 0.001 → 0.00001 Epoch: 50 → 250 Loss: Categorical → Binary Crossentropy	0.6222	0.9953	0.6055	0.5111
Config2	MaxPooling → AveragePooling Layer: 5	0.6129	0.8853	0.4679	0.6111
Config3	Dynamic learning rate	0.8443	0.6536	0.6330	0.5333
Config4	Original setup with grid search	0.8892	0.7990	0.7963	0.7829

Table 2: Performance comparison of different configurations.

5 Conclusions and Future Work

In this study, we evaluated the performance of Support Vector Machines (SVM) and Convolutional Neural Networks (CNN) for breast cancer classification using the CBIS-DDSM dataset. SVM models, despite their robustness on smaller feature sets, struggled to maintain accuracy on high-dimensional image data. CNNs, leveraging their capability for spatial feature extraction, achieved superior classification accuracy, particularly when combined with preprocessing techniques such as feature normalization and the inclusion of two-view mammographic images. This approach provided complementary perspectives, enhancing the detection of subtle abnormalities. While we initially considered using Transformers, we were unable to implement them due to the lack of detailed data preparation methods provided in the referenced papers. Instead, inspired by elements from both CNN and Transformer literature, we adopted an EfficientNet-based Convolutional Network, which demonstrated strong performance when optimized with proper configurations and preprocessing techniques, such as the Power Transformer for normalization.

With this experiment, I reevaluated different pattern recognition techniques used for image processing along with classical machine learning and modern deep learning architectures. The tests show how these models perform under different configurations and the possible ways to improve the outcomes. As part of future work, I plan to implement the swin transformer architecture (5) to generate classification results in a more effective and sophisticated manner and compare the performance with the previously performed experiments. In addition, I would like to test all of these methods with a different dataset such as VinDr-Mammo (6) and record the results to gain more insights on how these techniques play a part in providing with accurate and trustworthy results.

Future work should focus on enhancing datasets with diverse, multi-view images to maximize the benefits of two-view analysis. Developing hybrid models combining CNNs and Transformer-inspired architectures and integrating explainable techniques will ensure better performance and clinical applicability. Further optimization of computational efficiency and validation on external datasets will help improve robustness and scalability, advancing early breast cancer detection.

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